Project Proposal: "Albularyo's Apprentice"

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1. Introduction

A cozy merchant simulation PC game built in C++ with SFML, set in a pre-colonial-inspired Filipino village, where the player inherits an herbal shop after their albularyo master passes away.

The game involves brewing herbal potions, setting prices, managing supply and demand over time, and unlocking forgotten knowledge of rare ingredients. Only one potion is sold at a time, so timing is crucial.

Inspired by '*Lemonade Apocalypse*' by Juggler Interactive and '*Potion Craft: Alchemist Simulator*' by Niceplay Games.

2. Components of the game

2.1. Players

Number of Players**:** *Single-player*
Role: *Apprentice albularyo* (shop owner)

Other Characters:
- *Adventurers* deliver rare ingredients
- *Customers* purchase potions automatically

2.2. Game Environment

The game is set in a small herbal shop in a pre-colonial-inspired Filipino village. The wilderness exists conceptually but is not directly explorable. All external interactions (ingredient gathering, customers) occur as calculations.

2.3. Character Introduction and Narrative

"You are the apprentice of a respected albularyo.

After your master passes away, you inherit her shop. Initially, you continue brewing potions using leftover ingredients. However, over time, supplies begin to run low. You discover that you do not know how to gather ingredients the way your master did, and that several recipes require materials you cannot access.

Then one night, adventurers arrive looking for your master. They reveal that your master relied on them to gather rare ingredients from distant places. You agree to continue this arrangement."

2.4. Game Stats

The following are the stats of the player that change throughout gameplay:

1. Gold
2. Inventory of ingredients
3. Potion demand per time phase
4. Price setting
5. Demand-based sell chance

2.5. Game systems

The game has several systems, including:

1. Ingredient system
    a. There are two types of ingredients in the game, which are:
        i. Basic
            - Regenerates automatically at each time-phase
            - Ensures each player can always craft at least one potion
        ii. Rare
            - Delivered via adventurers
            - Must be paid per supply
            - Gate advanced potions
2. Potion system
    a. Uses *Base* + *Rare* ingredients for crafting
    b. Rare ingredients increase the value of the potion
3. Selling system
    a. Only one potion may be sold at a time

   b. One price slider
   c. Price directly affects demand
   d. Sales occur over time based on chance
      i. Chance is calculated by multiplying price, demand, and time.
4. Time system
   a. The game progresses in three phases:
      i. Sun time
      ii. Noon time
      iii. Night time
   b. Each phase has a fixed duration and modifies demand for each potion, as well as trigger regeneration of basic ingredients.
5. Tech tree system
   a. It is as follows:

Upgrades	*Effects*
Cat Companion	Increased spawn rate of butterflies
Bat Companion	Increased spawn rate of flowers
Murder of Crows	More customers at night
Awaken the Anito	Increased global potion demand
Hire more adventurers	Increased supply efficiency

2.6. Stages of the Game

The game has no traditional stages, but follows phases that are gated by certain conditions, as the following:

1. Phase 1 - Early Game
   - There are limited ingredients and only 1 craftable potion. The pricing system is introduced.
2. Phase 2 - Scarcity
   - Ingredients begin to run low and incomplete recipes are now visible. The player recognizes limitations.
3. Phase 3 - Adventurer Unlock (Narrative Event)
   - Triggered by previous gameplay conditions (Phase 2). The rare ingredients are introduced.
4. Phase 4 - Expansion
   - Multiple potions are now available and the player may now begin progressing through the tech tree. Optimizing gameplay becomes a priority.
5. Phase 5 - Endgame
   - The economy is now highly efficient, and the full tech tree is completed. The resources are stable.

2.6.1. Game Progression

The game progresses by:

1. Unlocking ingredient sources
2. Gaining access to more potions
3. Upgrading systems through the tech tree
4. Improving pricing efficiency

2.7. Goal of the Game

The goal of the game is to simply uncover the legacy of your master's work, as well as to stabilize the shop economy for maximum profit.

2.7.1. How to Play

To play the game, you must select a potion to 'feature' at the beginning of a time-phase. To gain profit, you may adjust the pricing using + and - button based on potion demand per second. You must manage your inventory supply via adventurers and unlocks in the tech tree.

2.7.2. Scoring system

There is no traditional score system, however progress may be tracked through total gold earned and unlocked systems in the tech tree.

2.7.3. How the Game Ends

The game ends once the shop reaches a final state, either through completion of the tech tree, or reaching a financial milestone (e.g. 5000 gold), where a final cutscene triggers and a message appears: "You've made your master proud."

3. Implementation needs

- The following are perceived data structures needed in the game:

```
C/C++
        struct Potion
        struct Ingredient
        enum class GameState
        struct TechNode (for tech tree)
        float sellChance
        int priceSliderValue
```

- The following show the perceived animation or graphics needed for the game:

Start screen:

Main screen (with instructions, Sun time):

Clean main screen (Noon time):

Dialogue sample (Moon time):

Potion making screen:

Skill/tech tree: